

Linux & SRE Interview Handbook

Command Cheat Sheet + Linux Fundamentals + Networking + Troubleshooting + Scenario Questions with Answers

Designed for: Linux, DevOps, SRE, Cloud, AWS, Kubernetes, Platform Engineering and Production Support interviews.

How to study: Memorize the cheat sheet first. Then practice saying the scenario answers out loud using a layer-by-layer troubleshooting flow instead of jumping randomly between commands.

Part I - Linux Command Cheat Sheet

The commands below are the short, high-frequency commands most useful in interviews and production troubleshooting.

System & OS

Command	Purpose / interview use
uname -a	Kernel, architecture, and system information.
uname -r	Kernel version.
cat /etc/os-release	Linux distribution/version.
hostname	Current hostname.
hostnamectl	Hostname and OS details on systemd systems.
uptime	Uptime and 1/5/15-minute load averages.
whoami	Current user.
id	UID, GID, and group memberships.
who	Logged-in users.
w	Logged-in users plus what they are doing.
date	Current date/time.
nproc	Number of available CPU cores.
lscpu	CPU architecture and topology.
env	Environment variables.
echo "\$PATH"	Executable search path.
command -v <cmd>	Resolve which executable/function/alias would run.

Files & Directories

Command	Purpose / interview use
pwd	Current directory.
ls -la	Detailed listing including hidden files.
mkdir -p a/b/c	Create nested directories.
touch file	Create empty file/update timestamp.
cp -r src dst	Copy directory recursively.
mv old new	Move or rename.
rm file	Delete file.
rm -rf dir	Delete recursively; dangerous.
file <name>	Identify file type.
stat <file>	Detailed metadata/inode/timestamps.
find /path -name "*.log"	Find files by name.
find / -type f -size +1G	Find files larger than 1 GB.
du -sh <dir>	Directory size.
df -h	Filesystem capacity.
df -i	Inode usage.
findmnt	Mounted filesystems.
mount	Mounted filesystems and mount options.
lsuf +L1	Deleted files still held open.

Permissions & Users

Command	Purpose / interview use
ls -l	View permissions/ownership.
chmod 755 file	Owner rwx; group/others r-x.
chmod 644 file	Owner rw; group/others r.
chmod +x script.sh	Make executable.
chown user:group file	Change owner and group.
sudo <cmd>	Run with elevated privileges.
visudo	Safely edit sudoers.
cat /etc/passwd	User account database.
groups <user>	User group memberships.
useradd <user>	Create user.
usermod -aG <group> <user>	Add user to supplemental group.
passwd <user>	Set/change password.

Processes, CPU & Memory

Command	Purpose / interview use
ps aux	Snapshot of all processes.
ps -ef	Full-format process list.
pgrep <name>	Find PIDs by process name.
top	Real-time process/CPU/memory view.
htop	Interactive top alternative if installed.
kill <PID>	Send SIGTERM by default.
kill -9 <PID>	Force SIGKILL.
pkill <name>	Signal processes by name.
free -h	Memory and swap usage.
ps aux --sort=-%cpu head	Top CPU consumers.
ps aux --sort=-%mem head	Top memory consumers.
vmstat 1	CPU, run queue, memory, paging, I/O every second.
iostat -xz 1	Extended disk I/O performance.
dmesg tail	Recent kernel messages.
journalctl -k	Kernel logs through systemd journal.
swapon --show	Configured/active swap.

Networking & DNS

Command	Purpose / interview use
ip addr	Interfaces and IP addresses.
ip route	Routing table/default gateway.
ip neigh	ARP/neighbor table.
ping <host>	ICMP reachability; failure does not prove service is down.
nslookup <host>	Basic DNS lookup.
nslookup <host> <dns>	Query a specific DNS server.
dig <host>	Detailed DNS query.
dig +short <host>	Compact DNS answer.

Command	Purpose / interview use
getent hosts <host>	Resolve using OS name-service rules, including /etc/hosts.
cat /etc/resolv.conf	Configured DNS resolvers/search domains.
ss -lntp	Listening TCP sockets and owning processes.
ss -tan	All TCP sockets, numeric output.
netstat -tulpn	Legacy: TCP/UDP listeners and processes.
lsof -i :8080	Process using port 8080.
nc -vz <host> <port>	Test raw TCP connectivity.
curl -v https://host	HTTP/TLS request with verbose diagnostics.
curl -I https://host	Fetch response headers.
curl -k https://host	Ignore TLS verification for diagnosis only.
traceroute <host>	Trace network hops.
tracpath <host>	Traceroute-style path test, often without root.

Logs & Services

Command	Purpose / interview use
systemctl status <svc>	Service status and recent log hints.
systemctl start <svc>	Start service.
systemctl stop <svc>	Stop service.
systemctl restart <svc>	Restart service.
systemctl enable <svc>	Enable at boot.
systemctl --failed	List failed units.
journalctl -u <svc>	Logs for one service.
journalctl -u <svc> -f	Follow service logs.
journalctl --since "1 hour ago"	Recent journal entries.
tail -f <log>	Follow text log.
tail -100 <log>	Last 100 lines.
grep -in "error" <log>	Case-insensitive search with line numbers.
grep -R "ERROR" /var/log	Recursive search.
awk	Field-oriented text processing.
sed	Stream editing/substitution.
sort uniq -c	Count repeated values/lines.

Shell, Redirection & Jobs

Command	Purpose / interview use
cmd > file	Overwrite stdout to file.
cmd >> file	Append stdout.
cmd 2> err.log	Redirect stderr.
cmd > out 2>&1	Redirect stdout + stderr.
cmd grep pattern	Pipe stdout to another command.
cmd &	Run in background.
jobs	Shell background jobs.
fg	Bring job to foreground.
nohup cmd &	Keep running after terminal disconnect.

Command	Purpose / interview use
source file	Execute file in current shell.
echo \$?	Previous command exit code.
set -euo pipefail	Common safer Bash scripting options.

Packages, Scheduling & Archives

Command	Purpose / interview use
apt install <pkg>	Install package on Debian/Ubuntu.
dnf install <pkg>	Install package on modern RHEL/Fedora.
yum install <pkg>	Older RHEL-family package manager.
crontab -l	List current user cron jobs.
crontab -e	Edit cron jobs.
tar -czvf file.tar.gz dir/	Create gzip-compressed tar archive.
tar -xzvf file.tar.gz	Extract gzip-compressed tar archive.
ln -s target link	Create symbolic link.
ln file hardlink	Create hard link.

Part II - SRE Troubleshooting Ladder

When an interviewer says, “The application cannot reach a service,” do not start by guessing. Walk the stack in order.

Step	What to check
1. Scope	Which users, hosts, AZs, pods, regions, or requests fail? When did it start?
2. Recent change	Deployment, config, DNS, IAM, certificate, route, firewall, scaling, maintenance?
3. DNS	nslookup / dig / getent. Does the name resolve to the expected IP?
4. Route	ip route. Is there a valid path/default route?
5. TCP	nc -vz host port. Timeout suggests drop/routing/firewall; refused suggests reachable host with no listener/reject.
6. TLS	curl -vk. Check certificate, SNI, protocol, trust chain.
7. HTTP	Interpret 401/403/404/429/5xx.
8. Identity	Credentials, IAM/RBAC, API token, role, policy.
9. App/dependency	Logs, CPU, memory, DB, downstream services, queues.
10. Mitigate + verify	Restore service first, then confirm from the real client path and prevent recurrence.

```
nslookup api.example.com
nc -vz api.example.com 443
curl -vk https://api.example.com
```

Interpretation: DNS fails -> DNS. DNS works but TCP times out -> route/firewall/SG/NACL. TCP works but TLS fails -> certificate/SNI. HTTP 401 -> authentication. HTTP 403 -> authorization/policy. HTTP 5xx -> application or dependency.

Part III - Full Linux Interview Question Bank

The remainder of the handbook expands the cheat sheet into interview-ready questions and concise answers. Commands are included where they add practical value.

Linux Fundamentals

1. What is Linux?

Linux is an open-source Unix-like kernel. A Linux distribution combines the kernel with user-space utilities, libraries, package managers, and applications.

2. What is the Linux kernel?

The privileged core of the OS. It manages CPU scheduling, memory, processes, networking, devices, filesystems, and system calls.

3. Linux vs Unix?

Unix is a family of operating systems with historical roots in AT&T; Unix. Linux is independently developed, Unix-like, and open source.

4. What is a Linux distribution?

A complete operating system built around the Linux kernel, such as Ubuntu, Debian, RHEL, Fedora, Rocky Linux, or Amazon Linux.

5. How do you identify the distribution?

Read /etc/os-release.

```
cat /etc/os-release
```

6. How do you check the kernel version?

Use `uname -r`; `uname -a` adds architecture and more system details.

```
uname -r
uname -a
```

7. How do you check the hostname?

Use `hostname` for the name or `hostnamectl` for broader system information.

```
hostname
hostnamectl
```

8. How do you check uptime and load?

`uptime` shows uptime, logged-in users, and 1/5/15-minute load averages.

```
uptime
```

9. How do you check the current user?

Use `whoami` or `id`.

```
whoami
id
```

10. How do you see logged-in users?

`who` shows sessions; `w` also shows what they are doing.

```
who
w
```

11. What is a process?

A running instance of a program with a PID, address space, file descriptors, credentials, and execution state.

12. What is PID 1?

The first userspace process started by the kernel. On most modern distributions it is `systemd`.

```
ps -p 1 -o pid,comm,args
```

13. What is a daemon?

A long-running background service, such as sshd, chronyd, or a web server.

14. What is a shell?

A command interpreter such as bash or zsh that launches programs, expands variables, manages pipelines, and supports scripting.

15. How do you check your shell?

Usually echo \$SHELL shows the login shell.

```
echo "$SHELL"
```

16. What is PATH?

A colon-separated list of directories the shell searches for executables.

17. Why might a command work for one user and fail for another?

Different PATH, permissions, shell profiles, aliases/functions, environment variables, package availability, or access controls.

18. What does source do?

Executes commands from a file in the current shell so environment changes persist.

```
. ~/.bashrc  
source ~/.bashrc
```

19. What is an environment variable?

A key/value pair inherited by child processes and commonly used for configuration such as PATH, HOME, or application settings.

20. How do you inspect environment variables?

Use env or printenv; echo a specific variable.

```
env  
printenv  
echo "$PATH"
```

Files, Directories & Links

21. What does pwd do?

Prints the current working directory.

```
pwd
```

22. What does ls -l show?

Long listing: permissions, link count, owner, group, size, timestamp, name.

```
ls -l
```

23. What does ls -la do?

Includes hidden dotfiles in the long listing.

```
ls -la
```

24. How do you create nested directories?

Use mkdir -p.

```
mkdir -p /tmp/app/logs
```

25. What does touch do?

Creates an empty file if missing or updates timestamps on an existing file.

```
touch file.txt
```

26. How do you copy a directory?

Use cp -r for recursive copy.

```
cp -r src dst
```

27. How do you move or rename a file?

mv performs both moves and renames.

```
mv old.txt new.txt
```

28. How do you delete a directory recursively?

rm -rf, but it is destructive and should be used carefully.

```
rm -rf directory
```

29. How do you identify a file type?

Use file; it inspects file contents/magic rather than just extension.

```
file filename
```

30. How do you find files by name?

Use find with -name.

```
find /var/log -name "*.log"
```

31. How do you find files modified in the last day?

Use find -mtime -1.

```
find /var/log -mtime -1
```

32. How do you find files larger than 1 GB?

Use find -size +1G.

```
find / -type f -size +1G 2>/dev/null
```

33. How do you find the size of a directory?

du -sh summarizes space used by files under a directory.

```
du -sh /var/log
```

34. What does df -h show?

Filesystem capacity, used space, free space, and mount points in human-readable units.

```
df -h
```

35. What is an inode?

A filesystem object storing metadata and pointers to file data. Directory entries map names to inode numbers.

36. How can “No space left on device” happen while GBs are free?

The filesystem may have no free inodes. Check df -i.

```
df -i
```

37. What is a mount?

An attachment of a filesystem to a directory in the Linux namespace.

38. How do you view mounts?

findmnt is usually clearer; mount also works.

```
findmnt  
mount
```

39. What is /etc/fstab?

Configuration describing filesystems that should be mounted, their mount points, types, and options.

40. What is a symbolic link?

A separate filesystem object containing a path to another file/directory. It can cross filesystems and can become broken.

```
ln -s target link
```

41. What is a hard link?

Another directory entry pointing to the same inode. It usually cannot cross filesystems.

```
ln file hardlink
```

42. What happens to a hard link if the original filename is deleted?

The data remains accessible through the other hard link as long as at least one link to the inode remains.

43. What does stat show?

Detailed file metadata: size, inode, mode, owner, timestamps, block count.

```
stat file
```

44. Why can df show a full filesystem while du shows much less?

A process may still have a deleted file open. The blocks remain allocated until the descriptor closes.

```
ls -l +L1
```

Permissions, Ownership & Users

45. Explain rwx permissions.

r = read, w = write, x = execute. They are defined separately for owner, group, and others.

46. What does chmod 755 mean?

Owner gets rwx; group and others get r-x.

```
chmod 755 script.sh
```

47. What does chmod 644 mean?

Owner gets rw-; group and others get r--.

```
chmod 644 file
```

48. How do you change ownership?

Use chown; user:group can change both.

```
chown user:group file
```

49. What does sudo do?

Runs a command under another security context, usually root, according to sudo policy.

50. What is root?

The superuser account with UID 0 and broad system privileges.

51. What is /etc/sudoers?

The main policy file controlling sudo access. Edit it with visudo to validate syntax.

52. Where are user accounts listed?

/etc/passwd contains account metadata; password hashes are generally in /etc/shadow.

53. What is /etc/shadow?

A root-readable file containing password hashes and password-aging information.

54. How do you see a user's groups?

Use groups or id.

```
groups username  
id username
```

55. How do you add a user to a supplemental group?

Use usermod -aG. The -a matters so existing supplemental groups are preserved.

```
sudo usermod -aG docker username
```

56. What is umask?

A mask that removes permission bits from default file/directory creation modes.

57. Why can a user read a file but not enter its directory?

Directory traversal requires execute permission on every parent directory.

58. What does the setuid bit do?

For an executable, it can run with the file owner's effective UID. It has security implications.

59. What does the sticky bit on /tmp do?

In a shared writable directory, it prevents users from deleting files they do not own.

60. What are ACLs?

Access Control Lists allow permissions for additional specific users/groups beyond traditional owner/group/other bits.

61. How do you inspect ACLs?

Use getfacl.

```
getfacl file
```

62. How do you set an ACL?

Use setfacl.

```
setfacl -m u:alice:r file
```

Processes, Signals & Jobs

63. How do you list processes?

ps aux or ps -ef are common snapshots.

```
ps aux  
ps -ef
```

64. How do you find a process by name?

Use pgrep; grep against ps also works but is less precise.

```
pgrep nginx
```

65. What does top show?

Real-time CPU, memory, load averages, tasks, and per-process utilization.

66. What is htop?

An interactive top alternative with easier navigation if installed.

67. How do you terminate a process gracefully?

Send SIGTERM using kill PID.

```
kill <PID>
```

68. What does kill -9 do?

Sends SIGKILL, which the process cannot catch or clean up after. Use only when graceful termination fails.

```
kill -9 <PID>
```

69. SIGTERM vs SIGKILL?

SIGTERM asks a process to exit cleanly. SIGKILL forces immediate termination by the kernel.

70. What is SIGHUP?

Historically terminal hangup; many daemons interpret it as reload configuration.

71. What is a zombie process?

A process that exited but whose parent has not reaped its exit status. It retains a process-table entry.

72. What is an orphan process?

A running process whose parent exited; it is adopted by another process such as systemd.

73. How do you run a command in the background?

Append &.

```
command &
```

74. What does jobs do?

Shows background/stopped jobs owned by the current shell.

75. How do you bring a job to the foreground?

Use fg, optionally with a job ID.

```
fg %1
```

76. What does nohup do?

Ignores SIGHUP so a process can continue after terminal disconnect; often combined with background execution.

```
nohup command > app.log 2>&1 &
```

77. What is nice/renice?

Utilities that influence CPU scheduling priority through niceness values.

78. What is a file descriptor?

A small integer used by a process to reference an open file/socket/pipe. Standard descriptors are 0 stdin, 1 stdout, 2 stderr.

79. How do you inspect open files for a process?

Use lsof -p PID.

```
lsof -p <PID>
```

80. How do you see a process tree?

Use pstree or ps --forest.

```
pstree -p  
ps -ef --forest
```

CPU, Load, Memory & Performance

81. What is load average?

The 1/5/15-minute average number of tasks runnable or in uninterruptible sleep, often including I/O wait. Compare it with CPU count and workload behavior.

82. Is load average the same as CPU utilization?

No. Load can be high because runnable tasks or tasks blocked in uninterruptible I/O are accumulating.

83. When is load “high”?

There is no universal threshold. Compare sustained load with CPU core count, run queue, latency, and I/O wait.

84. How do you find CPU count?

nproc or lscpu.

```
nproc  
lscpu
```

85. How do you find top CPU consumers?

Sort ps by %CPU or use top.

```
ps aux --sort=-%cpu | head
```

86. How do you check memory?

free -h summarizes RAM and swap.

```
free -h
```

87. Which is more useful: free or available memory?

Available is usually more useful because Linux uses otherwise idle RAM for cache and can reclaim much of it.

88. What is swap?

Disk-backed virtual memory used when pages are evicted from RAM. Heavy swap activity can cause severe latency.

89. How do you inspect swap?

swapon --show and free -h.

```
swapon --show  
free -h
```

90. What is OOM?

Out Of Memory. When memory cannot be satisfied, the kernel may invoke the OOM killer to terminate a process.

91. How do you detect OOM kills?

Check kernel/journal messages.

```
dmesg | grep -i oom
journalctl -k | grep -i oom
```

92. How do you find top memory consumers?

Sort ps by %MEM or use top.

```
ps aux --sort=-%mem | head
```

93. What does vmstat tell you?

Run queue, memory, swap, I/O, interrupts, context switches, and CPU breakdown.

94. What does iostat tell you?

CPU plus block-device I/O throughput, queueing, latency, and utilization.

95. What does high await mean in iostat?

I/O requests are waiting a long time, which can indicate storage latency or queueing.

96. What does high %util mean?

The device is busy for a high percentage of time; interpret with latency and workload characteristics.

97. What is iowait?

CPU time waiting for I/O completion. High iowait can indicate storage bottlenecks.

98. How would you investigate a slow server?

Check load/CPU, memory/swap, disk capacity/inodes, disk latency, network, application logs, and downstream dependencies before changing anything.

99. What is context switching?

The CPU switching execution from one thread/process to another. Excessive switching can hurt performance.

100. What is a load spike after a deployment likely to suggest?

Possibilities include increased traffic, a busy loop, excessive threads, changed query behavior, logging, retries, or blocked I/O. Correlate with the deployment and service metrics.

Networking, DNS, netstat, ss, lsof, nc & curl

101. How do you find IP addresses on Linux?

Use ip addr. ifconfig is legacy and may not be installed.

```
ip addr
```

102. Why is ip preferred over ifconfig?

iproute2 is the modern Linux networking toolkit and exposes routing, links, addresses, neighbors, and policy routing consistently.

103. How do you view the routing table?

Use ip route.

```
ip route
```

104. What is the default route?

The route used when no more specific destination route matches, typically via a gateway.

105. How do you test basic reachability?

ping can test ICMP reachability, but blocked ICMP does not prove the target service is unavailable.

```
ping example.com
```

106. What does nslookup do?

Queries DNS and shows the resolver response for names/records. It is useful for quick DNS verification.

```
nslookup example.com
```

107. How do you query a specific DNS server with nslookup?

Pass the server as the second argument.

```
nslookup example.com 8.8.8.8
```

108. If nslookup fails, what could be wrong?

Missing/wrong record, unreachable resolver, resolver configuration, private-zone/forwarding issue, network policy/firewall, or search-domain issue.

109. What is dig?

A detailed DNS diagnostic tool that exposes response status, flags, sections, TTLs, and server information.

```
dig example.com
```

110. How do you print only DNS answers?

Use dig +short.

```
dig +short example.com
```

111. What is getent hosts useful for?

It follows the OS name-service switch configuration, so it can include /etc/hosts and configured resolvers, unlike a raw DNS-only tool.

```
getent hosts example.com
```

112. What is /etc/resolv.conf?

Resolver configuration: nameservers, search domains, and resolver options. On some systems it is generated by systemd-resolved or NetworkManager.

113. What is /etc/hosts?

A local static mapping of hostnames to IP addresses that can take precedence depending on /etc/nsswitch.conf.

114. What does netstat do?

Legacy tool for sockets, listeners, routes, and interface statistics. Modern systems generally prefer ss and ip.

115. What does netstat -tulpn mean?

-t TCP, -u UDP, -l listening, -p process, -n numeric output.

```
sudo netstat -tulpn
```

116. Why might netstat not exist?

It is part of the older net-tools package and is often omitted by default.

117. What is ss?

Modern socket-inspection tool from iproute2 and the preferred replacement for many netstat use cases.

118. How do you show listening TCP ports with processes?

Use ss -lntp; process details may require privileges.

```
sudo ss -lntp
```

119. How do you see all TCP connections?

Use ss -tan.

```
ss -tan
```

120. How do you find whether port 443 is listening?

Filter ss output or use lsof.

```
sudo ss -lntp | grep :443
```

121. What does lsof -i :8080 do?

Shows processes with Internet sockets associated with port 8080.

```
sudo lsof -i :8080
```

122. What is nc/netcat used for?

A simple TCP/UDP utility commonly used to test whether a host:port is reachable.

```
nc -vz server.example.com 443
```

123. What does a successful nc connection prove?

DNS resolution (if a hostname was used), network path, and a TCP listener are working for that destination and port. It does not prove the application is functionally healthy.

124. What does TCP timeout usually suggest?

Packet drop, missing route, firewall/SG/NACL, unreachable network, dead target, or missing return path.

125. What does “connection refused” suggest?

The destination was reachable but no service accepted the connection on that port, or a firewall actively rejected it.

126. What does curl do?

Client for HTTP/HTTPS and other protocols. It is excellent for DNS/TCP/TLS/HTTP troubleshooting.

```
curl -v https://example.com
```

127. What does curl -I do?

Fetches response headers using a HEAD request where supported.

```
curl -I https://example.com
```

128. What does curl -k do?

Disables TLS certificate verification. Use only for troubleshooting, not as a real fix.

```
curl -vk https://example.com
```

129. How do you print only the HTTP status code?

Use curl write-out.

```
curl -o /dev/null -s -w "%{http_code}\n" https://example.com
```

130. What does traceroute show?

The series of routed hops toward a destination, subject to routers/firewalls responding to probe packets.

131. What is tracepath?

A traceroute-like utility that often works without elevated privileges and can help identify path/MTU behavior.

132. What is the TCP three-way handshake?

Client SYN -> server SYN-ACK -> client ACK.

133. What does LISTEN mean?

A TCP socket is waiting for incoming connections.

134. What does ESTABLISHED mean?

The TCP connection is open and data can be exchanged.

135. What is TIME_WAIT?

A TCP state after an active close that prevents delayed old packets from contaminating a later connection using the same tuple. Many TIME_WAIT sockets can be normal.

136. How do you count TCP states?

Summarize the first column of ss output.

```
ss -tan | awk '{print $1}' | sort | uniq -c
```

137. DNS uses UDP or TCP?

Most normal queries use UDP 53, but TCP 53 is also used for cases such as large responses and zone transfers.

138. What are common ports to know?

22 SSH, 25 SMTP, 53 DNS, 80 HTTP, 443 HTTPS, 3306 MySQL, 5432 PostgreSQL, 6379 Redis, 8080 common alternate HTTP/app port, 9200 Elasticsearch.

139. What is localhost?

The local host loopback address, commonly 127.0.0.1 for IPv4 and ::1 for IPv6.

140. Why can a service work locally but not remotely?

It may bind only to 127.0.0.1, a host firewall may block it, routing may be absent, or upstream network policy may block access.

141. How do you check what address a service is listening on?

Use `ss -lntp`. 127.0.0.1 means loopback-only; 0.0.0.0 means all IPv4 interfaces.

```
sudo ss -lntp
```

HTTP & Application Connectivity

142. What does HTTP 200 mean?

The request succeeded.

143. What do 301 and 302 mean?

Redirects. 301 is permanent; 302 is temporary in traditional semantics.

144. What is HTTP 400?

Bad Request: the server considers the request malformed or invalid.

145. What is HTTP 401?

Authentication is missing or invalid. Think credentials/token/session.

146. What is HTTP 403?

The request is understood but forbidden. Think authorization, policy, ACL, WAF, or business access rule.

147. What is HTTP 404?

The requested resource was not found, though some systems deliberately return 404 to hide resource existence.

148. What is HTTP 429?

Too Many Requests: rate limiting or quota enforcement.

149. What is HTTP 500?

Generic server-side application error.

150. What is HTTP 502?

Bad Gateway: a proxy/load balancer received an invalid/failed response from an upstream.

151. What is HTTP 503?

Service Unavailable: application or upstream is unavailable, overloaded, unhealthy, or intentionally out of service.

152. What is HTTP 504?

Gateway Timeout: a gateway/proxy did not receive a timely upstream response.

153. DNS works, TCP works, but HTTP returns 403. Where do you look?

Network reachability is mostly proven. Investigate authentication/authorization, IAM/RBAC, API keys, WAF/policy, headers, and application access rules.

154. curl returns connection refused. What do you check?

Check whether the target process is running and listening on the expected address/port, service binding, and active rejects.

```
ss -lntp
systemctl status <service>
```

155. curl times out. What do you check?

Route, host/network firewall, SG/NACL, proxy, load balancer, target reachability, and return path.

156. How would you troubleshoot a hostname end-to-end?

Resolve DNS, test TCP, test TLS/HTTP, then inspect identity and app logs.

```
nslookup api.example.com
nc -vz api.example.com 443
curl -vk https://api.example.com
```

SSH, Services, systemd & Logs

157. What is SSH?

Secure Shell: encrypted remote login and command execution, typically TCP port 22.

158. How do you connect with a private key?

Use ssh -i.

```
ssh -i key.pem user@server
```

159. Why does SSH reject a private key with loose permissions?

SSH protects private keys and may refuse keys accessible by other users. chmod 600 is common.

```
chmod 600 key.pem
```

160. Where are authorized SSH public keys stored?

Typically ~/.ssh/authorized_keys for the target user.

161. What is systemd?

The init/service manager on many modern distributions. It manages units, dependencies, lifecycle, logging integration, and boot targets.

162. How do you check a service?

Use systemctl status.

```
systemctl status nginx
```

163. How do you restart a service?

Use systemctl restart.

```
sudo systemctl restart nginx
```

164. How do you enable a service at boot?

systemctl enable creates the appropriate startup links/configuration.

```
sudo systemctl enable nginx
```

165. How do you list failed services?

Use systemctl --failed.

```
systemctl --failed
```

166. What is journalctl?

CLI for the systemd journal, aggregating structured logs from services, kernel, and system components.

167. How do you view one service's logs?

Use journalctl -u service.

```
journalctl -u nginx
```

168. How do you follow service logs?

Use -f.

```
journalctl -u nginx -f
```

169. How do you view logs from the last hour?

Use --since.

```
journalctl --since "1 hour ago"
```

170. What does dmesg show?

Kernel ring-buffer messages such as driver, hardware, filesystem, network, and OOM events.

171. How do you follow a text log?

Use tail -f.

```
tail -f /var/log/app.log
```

172. How do you see the last 100 lines?

tail -100.

```
tail -100 /var/log/app.log
```

173. A service will not start. What do you check first?

systemctl status and journalctl -u, then configuration syntax, permissions, port conflicts, missing dependencies, disk, credentials, and environment variables.

```
systemctl status <svc>  
journalctl -u <svc> --since "10 min ago"
```

174. What is a core dump?

A snapshot of a process memory/state at crash time used for postmortem debugging.

175. How do you inspect boot logs?

journalctl -b shows the current boot; -b -1 can show the previous boot.

```
journalctl -b
```

Text Processing, Pipes & Redirection

176. What does grep do?

Searches input for matching patterns.

```
grep "ERROR" app.log
```

177. How do you search case-insensitively with line numbers?

Use -i and -n.

```
grep -in "error" app.log
```

178. How do you recursively search a directory?

Use grep -R.

```
grep -R "ERROR" /var/log
```

179. What does awk do?

A pattern-scanning and field-processing language useful for selecting, aggregating, and transforming text.

180. What does sed do?

A stream editor commonly used for substitution, deletion, and simple text transformations.

```
sed 's/foo/bar/g' file
```

181. What does cut do?

Extracts delimited fields or character ranges.

```
cut -d: -f1 /etc/passwd
```

182. What does sort do?

Sorts input lines; options support numeric, reverse, key, and unique sorting.

183. What does uniq do?

Collapses adjacent duplicates, so it is commonly used after sort.

```
sort file | uniq -c
```

184. What does wc -l do?

Counts lines.

```
wc -l file
```

185. What is a pipe?

It connects stdout of one process to stdin of another.

```
ps aux | grep nginx
```

186. What does > do?

Redirects stdout to a file and truncates/overwrites the file.

187. What does >> do?

Appends stdout to a file.

188. What are stdin/stdout/stderr descriptors?

0 stdin, 1 stdout, 2 stderr.

189. How do you redirect stderr?

Use 2>.

```
command 2> error.log
```

190. How do you redirect stdout and stderr together?

Use > file 2>&1 or Bash &>.

```
command > output.log 2>&1
```

191. What is /dev/null?

A special device that discards anything written to it.

```
command > /dev/null 2>&1
```

192. What does tee do?

Copies stdin to stdout and one or more files, useful when you want to view and save output.

```
command | tee output.log
```

Package Management, Cron, Archives & Shell Scripting

193. What package manager is common on Debian/Ubuntu?

apt/apt-get, backed by dpkg.

194. What package manager is common on modern RHEL/Fedora?

dnf, backed by RPM.

195. What is yum?

Older RHEL-family package manager; on many newer systems it is a compatibility layer for dnf.

196. How do you install curl on Ubuntu?

sudo apt install curl.

```
sudo apt update && sudo apt install curl
```

197. How do you install curl on RHEL/Fedora?

sudo dnf install curl.

```
sudo dnf install curl
```

198. What is cron?

A time-based job scheduler for recurring commands.

199. How do you view your cron jobs?

crontab -l.

```
crontab -l
```

200. What is the cron field order?

minute hour day-of-month month day-of-week command.

201. What does “0 2 * * *” mean?

Run daily at 02:00.

202. How do you create a gzip tar archive?

tar -czvf.

```
tar -czvf backup.tar.gz directory/
```

203. How do you extract it?

```
tar -xzvf.
```

```
tar -xzvf backup.tar.gz
```

204. What is a shebang?

The first line such as `#!/bin/bash` that tells the OS which interpreter to use for an executable script.

205. How do you make a script executable?

```
chmod +x script.sh.
```

```
chmod +x script.sh
```

206. What is \$1 in Bash?

The first positional argument.

207. What is \$#?

The number of positional arguments.

208. What is "\$@"?

All positional arguments preserved as separate quoted arguments.

209. What is \$?

Exit status of the previous command. 0 conventionally means success.

210. What does set -e do?

Requests the shell to exit in many cases when a command fails; behavior has caveats in conditionals and pipelines.

211. What does set -u do?

Treats expansion of unset variables as an error.

212. What does set -o pipefail do?

Makes a pipeline fail if any command in the pipeline fails rather than only considering the final command.

213. Why use set -euo pipefail?

It can make scripts fail faster on common errors, but production scripts still need explicit error handling for important operations.

214. How do you write an if test for a file?

Use `[ -f file ]`.

```
if [ -f file.txt ]; then echo "exists"; fi
```

215. How do you write a basic loop?

Use `for/while` constructs.

```
for x in a b c; do echo "$x"; done
```

Boot, /proc, /sys & Kernel Concepts

216. What is the simplified Linux boot sequence?

BIOS/UEFI -> bootloader -> Linux kernel -> initramfs -> PID 1/systemd -> services/targets.

217. What is GRUB?

A common bootloader that loads the selected kernel and initramfs and passes kernel parameters.

218. What is /proc?

A virtual filesystem exposing process and kernel state, such as `/proc/cpuinfo`, `/proc/meminfo`, and `/proc/`.

219. What is /sys?

A virtual filesystem exposing kernel devices, drivers, attributes, and parts of the device model.

220. What is a system call?

A controlled interface by which a userspace process requests kernel operations such as `read`, `write`, `open`, `fork`, or `network calls`.

221. What is user space vs kernel space?

User space runs normal applications with restricted privileges; kernel space runs privileged kernel code with direct hardware/system control.

222. What is initramfs?

An initial RAM filesystem used during early boot to load drivers/tools needed to find and mount the real root filesystem.

223. What is a kernel module?

Loadable kernel code used for drivers or optional kernel functionality.

```
lsmod
```

224. How do you see loaded kernel modules?

Use lsmod.

```
lsmod
```

225. What does sysctl do?

Reads/writes many runtime kernel parameters exposed through /proc/sys.

```
sysctl -a | head
```

Production Troubleshooting Scenarios

226. A server is unreachable. What is your troubleshooting order?

Confirm scope and recent changes; check your own interface/IP, route/default gateway, DNS, TCP port, path, destination listener, and firewalls. Use evidence before making changes.

```
ip addr
ip route
nslookup <host>
nc -vz <host> <port>
traceroute <host>
```

227. DNS works but the connection times out. What does that mean?

Name resolution is not the primary problem. Investigate routes, security groups/NACLs/firewalls, target availability, proxies/load balancers, and the return path.

228. DNS works and TCP works but the app gets HTTP 403. What does that suggest?

Connectivity is good enough to reach the application or gateway. Investigate identity, authorization, IAM/RBAC, token scopes, WAF, ACLs, or application policy.

229. A server CPU is at 100%. What do you do?

Identify top consumers, inspect whether load is user CPU, system CPU, iowait, or steal, correlate with traffic and recent changes, and mitigate safely by fixing the runaway process or adding capacity.

```
top
ps aux --sort=-%cpu | head
vmstat 1
```

230. Memory is almost full. What do you do?

Check available memory, top consumers, swap activity, OOM logs, and whether usage is cache vs process growth. Investigate leaks or workload changes before restarting blindly.

```
free -h
ps aux --sort=-%mem | head
journalctl -k | grep -i oom
```

231. Disk is 100% full. What do you do?

Check filesystem and inode usage, find large directories/files, check logs and deleted-open files, then clean or expand safely.

```
df -h
df -i
du -xhd1 / | sort -h
lsdf +L1
```

232. Port 8080 is already in use. How do you find the process?

Use ss or lsof.

```
sudo ss -lntp | grep :8080
sudo lsof -i :8080
```

233. A service will not start. What do you check?

Status and logs first, then config, permissions, ports, dependencies, disk, credentials, environment, and recent changes.

```
systemctl status <service>
journalctl -u <service>
```

234. A process keeps crashing. What do you investigate?

Application and kernel logs, exit code/signal, OOM, missing libraries/config, permissions, dependency failures, bad inputs, and core dumps.

235. The app is slow but CPU is low. What next?

Check I/O wait, disk latency, memory/swap, network latency, connection pools, database/query latency, locks, queues, and downstream dependencies.

236. The app is slow only for one region/AZ. What does that suggest?

A localized network path, capacity, dependency, routing, AZ resource, load-balancer target, or data-store issue. Compare metrics and topology across healthy vs unhealthy locations.

237. A process was killed with no app error. Where do you look?

Kernel logs for OOM or signals, service manager logs, container/runtime events, and external supervisors.

238. You see many TIME_WAIT sockets. Is that an outage?

Not by itself. It is expected with many short-lived TCP connections. Investigate only if ephemeral ports, connection churn, or latency are actually causing impact.

239. You see many CLOSE_WAIT sockets. What does that suggest?

The remote peer closed, but the local application has not closed its socket; this can indicate an application connection-handling leak.

240. A hostname resolves to the wrong IP. What do you inspect?

DNS record/TTL, resolver being queried, search domains, /etc/hosts, split-horizon/private zones, caches, and environment-specific DNS configuration.

241. A service resolves locally but not inside a container/pod. What do you compare?

Container/pod resolver configuration, DNS service/CoreDNS, namespace network policy, search domains, and whether the host uses a different resolver path.

242. An application says “connection refused” after deployment. What are likely causes?

Process failed to start, wrong port, wrong bind address, readiness issue, misconfigured service/target, or listener is not yet ready.

243. An application times out after a firewall change. How do you prove where it fails?

Resolve name, test TCP from the real source, inspect local route, capture/flow logs if available, check firewall rules and return path, compare a known-good source.

244. How do you troubleshoot intermittent failures?

Correlate timestamps across logs/metrics, look for load balancing, DNS TTL, conntrack/NAT exhaustion, resource saturation, autoscaling, dependency flaps, and packet loss. Preserve evidence before restarting.

245. How do you handle an incident when you do not know the root cause yet?

State known impact/scope, mitigate user impact, collect evidence, form and test hypotheses, communicate updates, verify recovery, then complete root-cause and prevention work.

SRE & Reliability Interview Questions

246. What is SRE?

Site Reliability Engineering applies software engineering and operational practices to keep services reliable, scalable, observable, and maintainable.

247. What is an SLI?

A Service Level Indicator is a measured signal of service behavior, such as availability, latency, correctness, throughput, or freshness.

248. What is an SLO?

A Service Level Objective is a target for an SLI over a time window, such as 99.9% successful requests per 30 days.

249. What is an SLA?

A Service Level Agreement is an external or contractual commitment and may include consequences for missing the target.

250. What is an error budget?

The amount of unreliability allowed by an SLO. It creates a framework for balancing feature velocity and reliability work.

251. What is MTTR?

Mean Time To Restore/Recover, depending on organizational definition: how long service restoration takes after an incident.

252. What is MTTD?

Mean Time To Detect: how quickly incidents are detected after they begin.

253. What is toil?

Manual, repetitive, automatable operational work that scales linearly with service growth and provides little enduring value.

254. What makes a good alert?

It should be actionable, tied to user/service impact, have clear ownership/runbook context, and avoid unnecessary noise.

255. What is alert fatigue?

Responders receive so many low-value/noisy alerts that meaningful alerts are easier to miss.

256. Symptom-based vs cause-based alerting?

Symptom alerts focus on user-visible failures such as high error rate or latency; cause alerts focus on components like CPU. Symptom alerts usually map more directly to incidents.

257. What is a runbook?

A documented operational procedure containing checks, decision points, mitigations, validation, ownership, and escalation information.

258. What is a postmortem?

A structured incident review documenting impact, timeline, contributing factors/root cause, response, and concrete prevention actions.

259. Why avoid blame in postmortems?

Blameless analysis focuses on system/process conditions so the organization can learn and reduce recurrence rather than hide mistakes.

260. What should you do first during a major incident?

Establish ownership/coordination, understand impact and blast radius, mitigate user harm, preserve evidence, and communicate clearly.

261. What is a rollback vs roll-forward?

Rollback restores a known prior version; roll-forward applies a corrective change. Choose the safest fastest path based on confidence and reversibility.

262. Why is restarting not always a root-cause fix?

It may clear symptoms while destroying evidence and allowing recurrence. Restart can be a valid mitigation, but root cause and prevention still need investigation.

263. What is graceful degradation?

A system continues providing reduced but useful functionality when a dependency or feature is unavailable.

264. What is a circuit breaker?

A resilience pattern that stops repeated calls to an unhealthy dependency temporarily, limiting cascading failures.

265. What is backpressure?

A mechanism that slows or rejects incoming work when consumers cannot keep up, preventing unbounded queue/resource growth.

266. What is exponential backoff?

Retry delays increase after failures, often with jitter, to reduce retry storms and give dependencies time to recover.

267. Why add jitter to retries?

It spreads retries over time so many clients do not retry simultaneously and create a thundering herd.

268. What is idempotency?

Repeating the same operation produces the same intended effect, important for safe retries in distributed systems.

269. How would you reduce MTTR?

Better detection, actionable alerts, strong dashboards/logs/traces, clear runbooks, ownership, safe automated diagnostics, rollback paths, and incident practice.

270. How do you prioritize reliability work?

Use user impact, SLO/error-budget burn, recurrence frequency, severity, operational toil, and risk rather than only infrastructure neatness.

Rapid-Fire Interview Questions

271. How do you check DNS?

nslookup, dig, or getent hosts.

```
nslookup host
dig host
getent hosts host
```

272. How do you test a TCP port?

nc -vz host port.

```
nc -vz host 443
```

273. How do you inspect listening ports?

ss -lntp.

```
sudo ss -lntp
```

274. How do you check routes?

ip route.

```
ip route
```

275. How do you check IP addresses?

ip addr.

```
ip addr
```

276. How do you check CPU?

top, ps sorting, vmstat, or mpstat if installed.

```
top
```

277. How do you check memory?

free -h plus process-level inspection.

```
free -h
```

278. How do you check disk capacity?

df -h.

```
df -h
```

279. How do you check inodes?

df -i.

```
df -i
```

280. How do you check open files?

lsof.

```
lsof
```

281. How do you check a service?

systemctl status.

```
systemctl status <svc>
```

282. How do you view service logs?

journalctl -u.

```
journalctl -u <svc>
```

283. How do you follow a log?

tail -f or journalctl -f.

```
tail -f app.log
```

284. How do you find a process?

pgrep or ps.

```
pgrep <name>
```

285. How do you find the process using port 8080?

lsof -i :8080 or ss -lntp.

```
sudo lsof -i :8080
```

286. How do you check load?

uptime or top.

```
uptime
```

287. How do you inspect kernel errors?

dmesg or journalctl -k.

```
journalctl -k
```

288. How do you list environment variables?

env or printenv.

```
env
```

289. How do you check if a secret variable is set without printing it?

Test its length and print only SET/NOT_SET.

```
[ -n "$PASSWORD" ] && echo SET || echo NOT_SET
```

290. How do you determine which executable runs?

command -v is portable for shell resolution; which is also common.

```
command -v python
```

291. How do you check filesystem mounts?

findmnt.

```
findmnt
```

292. How do you see recent failed logins?

Depending on distro/auditing, use journalctl for sshd or lastb if enabled.

```
journalctl -u sshd --since today
```

293. How do you check the default gateway?

Look for default via in ip route.

```
ip route | grep ^default
```

294. How do you check if a port is bound only to localhost?

Inspect the local address in ss -lntp.

```
ss -lntp
```

295. How do you inspect DNS resolver configuration?

Read /etc/resolv.conf and, on systemd-resolved systems, resolvectl status.

```
cat /etc/resolv.conf  
resolvectl status
```

296. How do you flush DNS cache?

It depends on resolver. On systemd-resolved: resolvectl flush-caches.

```
sudo resolvectl flush-caches
```

297. How do you check a TLS certificate from the command line?

openssl s_client can inspect the handshake/certificate chain.

```
openssl s_client -connect example.com:443 -servername example.com
```

298. How do you see disk devices and partitions?

lsblk.

```
lsblk
```

299. How do you see filesystem types?

lsblk -f or df -T.

```
lsblk -f  
df -T
```

300. How do you see network interface counters?

ip -s link.

```
ip -s link
```

301. How do you see ARP/neighbor entries?

ip neigh.

```
ip neigh
```

302. How do you list systemd units?

systemctl list-units.

```
systemctl list-units --type=service
```

303. How do you reload systemd unit files after editing them?

systemctl daemon-reload.

```
sudo systemctl daemon-reload
```

304. How do you reload a service without full restart if supported?

systemctl reload .

```
sudo systemctl reload nginx
```

305. How do you verify a config before restarting nginx?

nginx -t.

```
sudo nginx -t
```

306. How do you test whether an HTTP service returns 200?

curl with status code output.

```
curl -sS -o /dev/null -w "%{http_code}\n" https://host
```

307. How do you time an HTTP request?

curl can expose connect, TLS, start-transfer, and total time.

```
curl -sS -o /dev/null -w "connect=%{time_connect} start=%{time_start}transfer} total=%{time_total}\n" https://host
```

308. How do you capture packets?

tcpdump, usually requiring privileges.

```
sudo tcpdump -i any host <ip> and port 443
```

309. When would you use tcpdump?

When higher-level evidence is insufficient and you need to confirm whether packets leave/arrive, handshakes occur, retransmissions happen, or DNS traffic is visible.

310. How do you inspect process environment?

/proc//environ can expose it, but be careful with secrets. Prefer process-specific safe inspection.

```
tr "\0" "\n" < /proc/<pid>/environ
```

311. How do you inspect process command line?

Read ps or /proc//cmdline.

```
ps -p <pid> -o args=
```

312. How do you check current open-file limits?

ulimit -n for the shell; process limits can be read from /proc//limits.

```
ulimit -n
```

313. What can “too many open files” mean?

The process/system file-descriptor limit is exhausted, often due to connection/file leaks or undersized limits.

314. How do you see a process’s open FD count?

Count entries under /proc//fd.

```
ls /proc/<pid>/fd | wc -l
```

315. How do you check time synchronization?

timedatectl and chronyc/ntpq depending on the time service.

```
timedatectl
```

316. Why does time synchronization matter?

TLS, distributed logs, authentication tokens, databases, and incident timelines depend on reasonably synchronized clocks.

317. What is localhost DNS vs external DNS difference?

Local resolution can be influenced by /etc/hosts, NSS modules, local caches, VPN DNS, and split-horizon configuration; external resolvers may return different answers.

Part IV - What to Memorize Before the Interview

- **Networking:** nslookup/dig/getent -> ip route -> nc -> curl -> ss/lsof -> logs.
- **Performance:** uptime/top -> free -h -> vmstat -> df -h/df -i -> iostat -> app/dependency metrics.
- **Services:** systemctl status -> journalctl -u -> check port/config/permissions/dependencies.
- **Disk:** df -h, df -i, du, find large files, lsof +L1.
- **Process:** ps/pgrep/top, SIGTERM before SIGKILL, check OOM if unexplained death.
- **HTTP:** 401 auth, 403 authorization/policy, 429 rate limit, 502 bad upstream, 503 unavailable, 504 upstream timeout.
- **SRE answer structure:** impact -> scope -> recent change -> evidence -> mitigation -> verification -> prevention.

Total detailed interview questions in this PDF: 317.

Best interview habit: Explain what a command proves. For example, do not just say “I would run nc.” Say “I would use nc to test whether the TCP path and listener are reachable; a timeout points me toward routing/firewall/return-path issues, while connection refused tells me the destination is reachable but not accepting that port.”